

Project Title

Indian Renewable Energy Data Analytics

Description

This project analyses renewable energy generation across Indian states using daily data from 2020 to 2024. The analysis focuses on solar energy, wind energy, other renewable energy, total renewable generation, generation performance, and installed capacity. Python was used for data cleaning and Exploratory Data Analysis (EDA), while Power BI was used to create interactive dashboards and communicate key findings.

India's renewable energy sector generates large amounts of data across different states and time periods. This project aims to analyse that data to understand renewable generation trends, compare states and energy sources, identify high and low-performing regions, and present the findings through clear and interactive visualizations.

Getting Started

Dependencies

The following tools and libraries are required:

- Python 3.x
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Power BI Desktop

Installing

1. Download this project repository from GitHub.
2. Open the project in **Google Colab** or **Jupyter Notebook**.
3. Upload the dataset: `processed_daily_state.csv`
4. Install the required Python libraries if they are not already available
5. No major modifications are required in the dataset files before running the project.

Executing program

1. Open the Python notebook in Google Colab or Jupyter Notebook.
2. Import the required libraries:
`import pandas as pd`
`import numpy as np`
`import matplotlib.pyplot as plt`
`import seaborn as sns`
3. Load the dataset:
`df = pd.read_csv("processed_daily_state.csv")`
4. Check the structure of the dataset:
`df.head()`
`df.info()`
`df.shape`
5. Perform data cleaning and preprocessing.

6. Check for:
 - o Missing values
 - o Duplicate records
 - o Incorrect data types
 - o Negative or invalid values
 - o Inconsistent data
7. Create the required calculated columns and prepare the cleaned dataset.
8. Perform Exploratory Data Analysis (EDA) using Python.
9. Create visualizations to analyze:
 - o State-wise generation
 - o Solar energy
 - o Wind energy
 - o Other renewable energy
 - o Total renewable energy
 - o Capacity
 - o Monthly trends
 - o Yearly trends
 - o State and regional comparisons
10. Export the cleaned dataset for Power BI.
11. Open **Power BI Desktop** and import the cleaned dataset.
12. Create the required:
 - KPI cards
 - Bar charts
 - Line charts
 - Scatter plots
 - Comparison charts
 - State-wise analysis
13. Add filters and slicers to make the dashboard interactive.
14. Analyze the dashboard and identify key business insights.

Help

- If Python cannot find the dataset, make sure the CSV file is uploaded to Google Colab or placed in the correct project folder.
- Make sure Matplotlib and Seaborn are imported correctly.
- Use the cleaned dataset for dashboard creation.

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Version History

0.2

- Added data cleaning and preprocessing.
- Added Exploratory Data Analysis.
- Added renewable energy visualizations.
- Added Power BI interactive dashboards.
- Added KPI and state-wise analysis.
- Added dashboard insights and comparisons.

0.1

- Initial project release.
- Added renewable energy dataset.
- Performed initial data exploration.

`## License`

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`## Acknowledgments`

- Thanks to the data source and contributors who made the renewable energy data available.
- Python and its open-source data analysis libraries.
- Microsoft Power BI for dashboard development.
- Google Colab for the Python analysis environment.
- Open-source community resources that helped during project development.